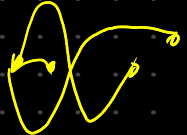


object: - drawing rule from theorem 3.1, the case where we might have 

How does the proof of theorem 3.1 work?

"if we encounter an odd pair, we can either make it cross evenly or [we are in a situation that has been excluded by lemma 3.3]"

we need to focus on this part

Lemma 3.3: suppose G is a min counterexample to th. 3.1

connected $\leftarrow$ G $\rightarrow$ not vertex min. $\leftarrow$

with H $\downarrow$ $V(H) \cup \{a, b\} \neq \emptyset$

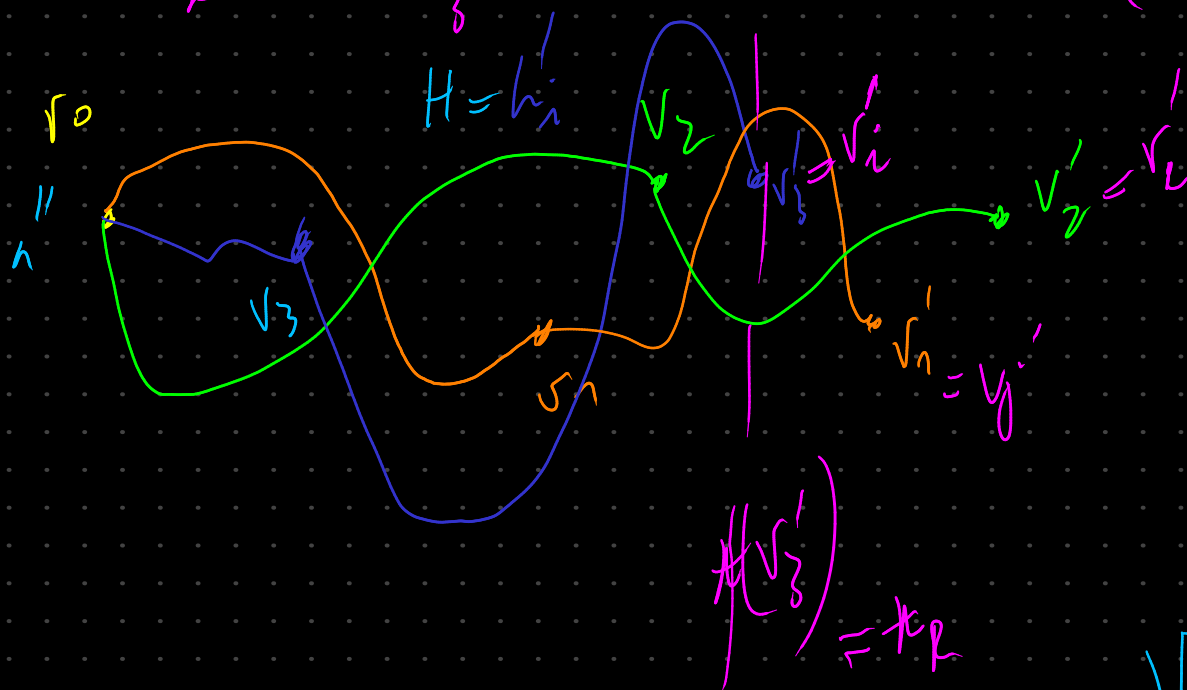
basically we suppose that G is a min counterexample that have  with all the graph settings we could

have, and prove that also this graph can be a monotone drawing (contradiction)

Input, Graph h with n odd crossings, Can we make it π -monotone

We pick an odd crossing pair:
with v_0 the common endpoint of the pair of edges.
we take h_0 subgraph of h with $\pi(v) \geq \pi(v_0)$

Case 1: v_1, v_2, v_3 are in different comp
- Case 1.1: v_3' is the smallest ($i, i_k = (3, 1, 2)$)



Lemma 3.4

(no height of v_1', v_2', v_3' on the left)

Lemma 3.3 (iii)

I think the focus should be on Lemma 3.3

defining what can be turned to a π -plane embedding or not.

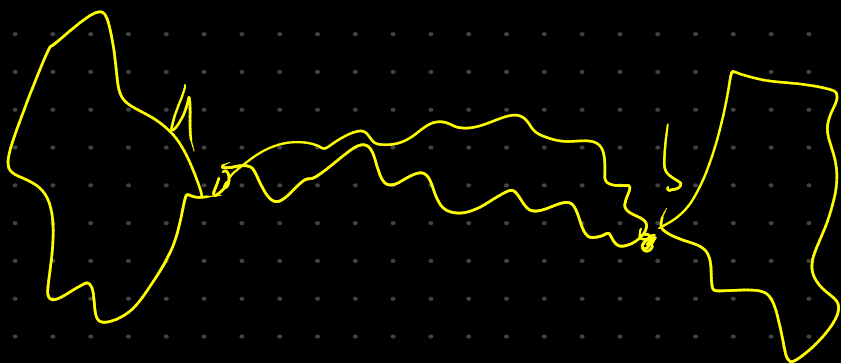
iii) h has a cut-vertex a and $h - \{a\}$ has a comp. H such that $\pi(a) < \pi(v) \forall v \in V(H)$. Then H has only one vertex b , and h has no edge ac with $\pi(b) < \pi(c)$. Also h has no con. subgraph

$$H' \neq \emptyset \text{ s.t. } \begin{cases} \pi(v) \in]\pi(a), \pi(b)[\quad \forall v \in V(H') \\ a \notin N(H'), \quad N(H') \neq \{a\} \\ \pi(v) > \pi(b) \quad \forall v \in N(H') - \{a\} \end{cases}$$

let's consider $H \in \mathcal{C}$ of $h - \{a\}$, $\pi(a) < \pi(v) \forall v \in H$.

let b largest π -value in H .

- If $|H| > 7 \xrightarrow{\text{case (ii)}} H := H - b$



base of ii) no, connected $H \subset G$

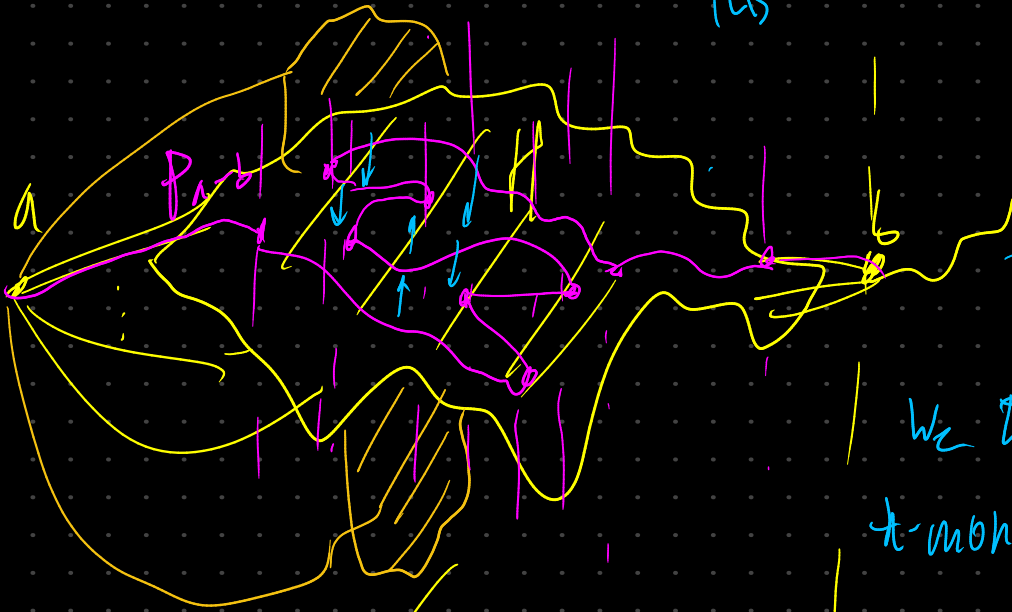
$$a, b \in V(G) - V(H)$$

$$\{ \kappa(a) < \kappa(H) < \kappa(b) \mid \forall v \in V(H) \}$$

$$N(H) = \{a, b\}$$

Why does this matter?

$$(V(G) - (V(H) \cup \{a, b\})) \neq \emptyset$$



$$G[V(H) \cup \{a, b\}]$$



We have $G - V(H)$
 κ -monotone drawing
 adding κ -monotone edge
 with same priority crossing
 as P (even) $\Rightarrow$

$G - V(H) \cup \{a, b\}$ is
 H -T drawing and # of
 vertices is smaller than
 $G \Rightarrow \kappa$ -monotone embedding.

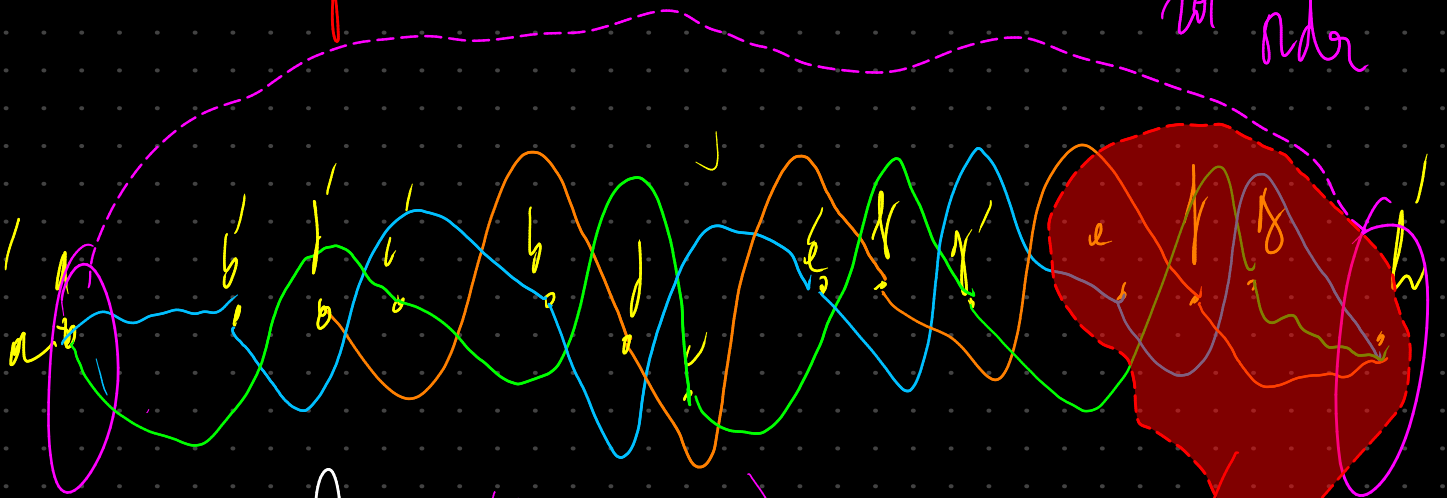
flatten

$$G[V(H) \cup \{a, b\}]$$

and draw it next to $\{a, b\}$

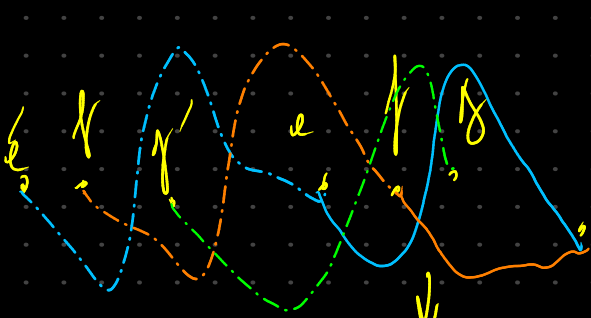
importance of lemma 2.3

same not order

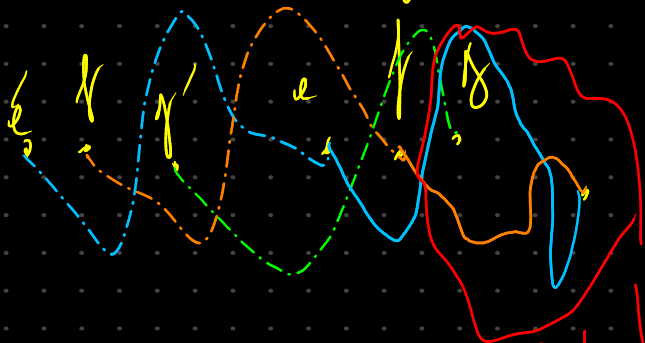


$\mathcal{I}$

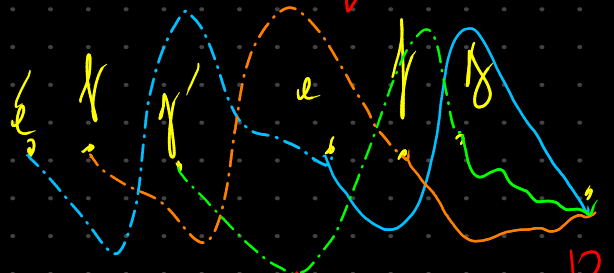
(Theorem 3.1)
Case 2)



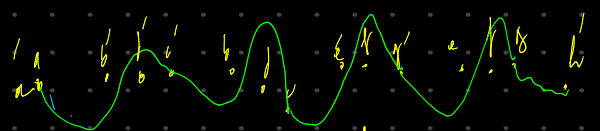
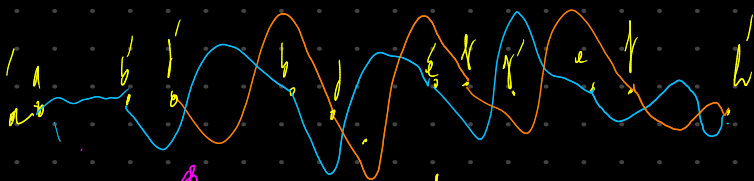
we can look at it like



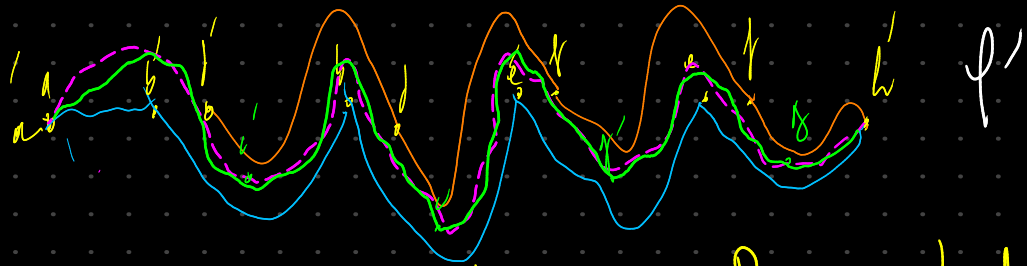
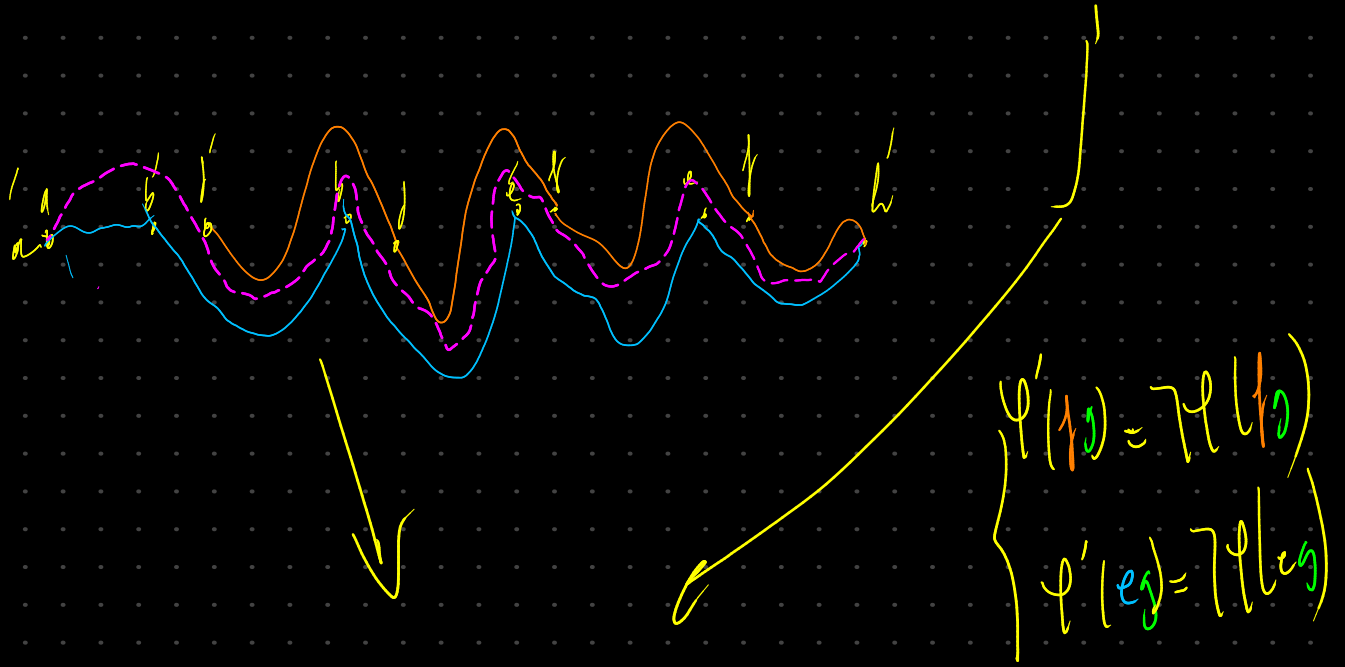
back to lemma 2.3



odd crossing with
in edge in between
in the rotation
(Theorem 3.1)
cases

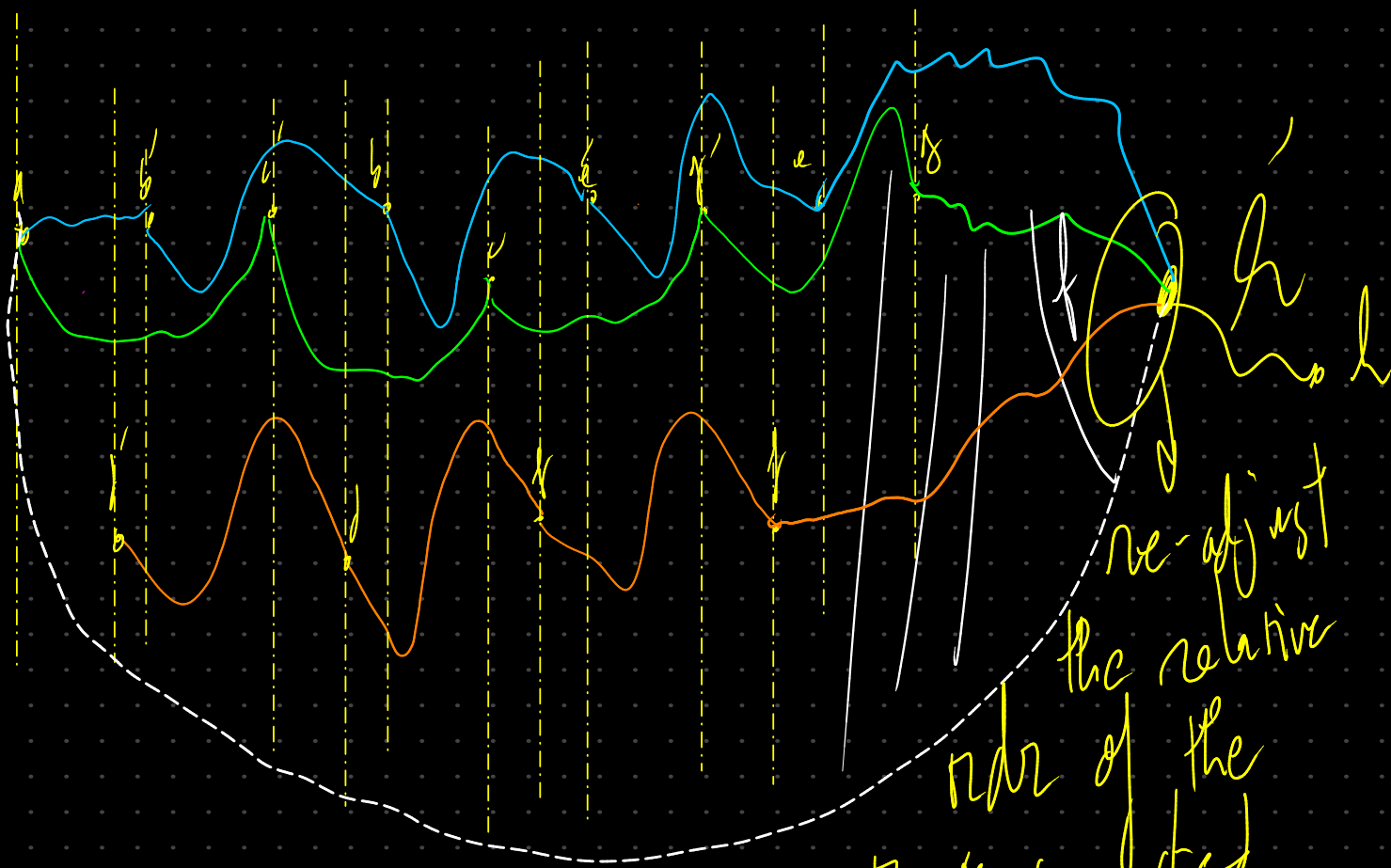
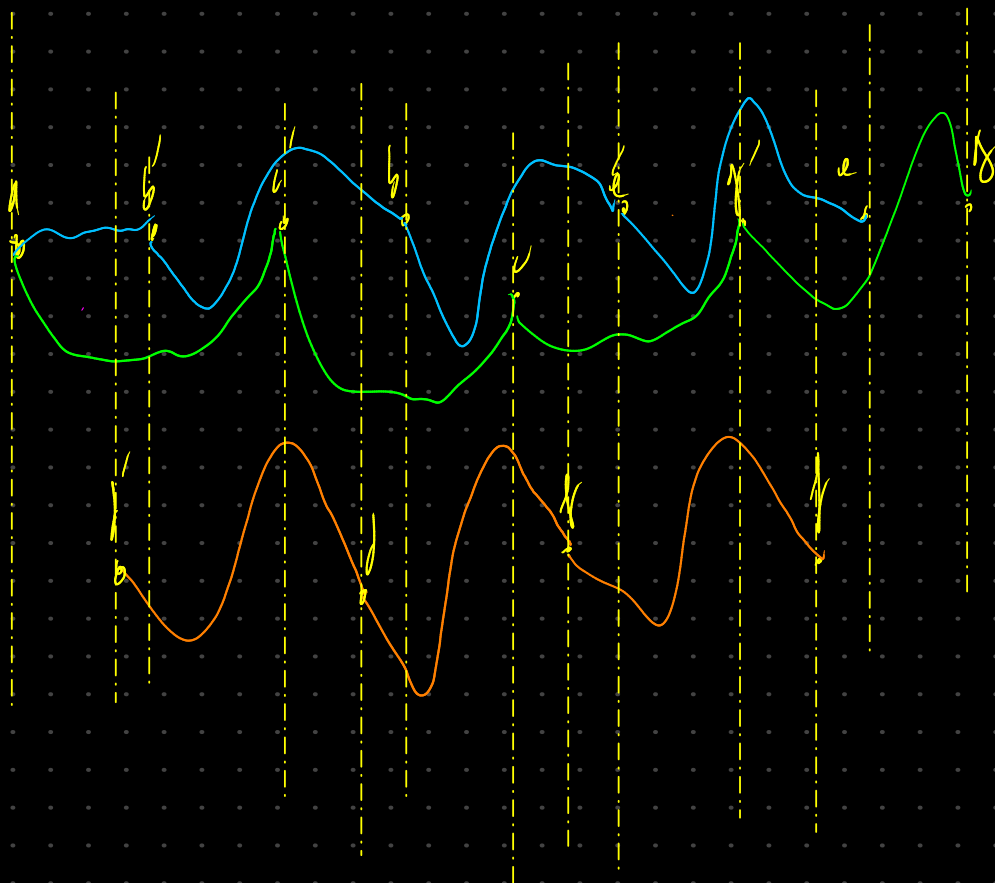


Weak
lemma full



if we start the construction of the embedding from the right we end up with this case

if we start from the left of h



re-adjust
the relative
range of the
comp to the ndr
in the rotation of h